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Managed Pressure Drilling (MPD) and Continuous Circulating System (CCS) Unlock UAE's Deepest High-Temperature Exploration Well

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Abstract

Exploration wells are drilled to discover and evaluate new hydrocarbon reservoirs in unexplored areas; it plays a critical role in the growth and sustainability of an Oil Company. The continuous effort and investment in adding potential reserves are essential to ensuring future profitability and operational success of the company.

Worldwide the drilling exploration wells represents a significant challenge for operators, as it typically involves targeting reserve areas that were previously inaccessible or not viable to reach with the existing technology.

That is the case of an exploration well in the United Arab Emirates that aimed to reach one of the deepest exploration targets in the Middle East, the formations in Permian basin.

The well faced typical exploration challenges, bottom hole pressure uncertainties and extremely sub surface temperature. Planning this ambitious project began in 2019 but experienced several delays until 2024 became a reality.

This paper documents the successful application of Managed Pressure Drilling (MPD) and Continuous Circulating System (CCS) technologies in the deepest onshore exploration well in the UAE, reaching over 19,000 ft Total Vertical Depth (TVD) and encountering static bottomhole temperatures above 450°F. MPD facilitated Early Kick Detection (EKD) and CCS ensured bottom hole temperature stability. The combined system allowed the operator to achieve the target formation and data acquisition without requiring a contingency liner, logging or coring failures, or encountering any influx or loss events. The success of this project validates the synergy of MPD and CCS in managing the complex dynamic environment of high-temperature wells.

Introduction

In exploration wells, conventional drilling methods often struggle to manage narrow pressure margins—particularly as bottomhole temperatures begin to rise, increasing the risk of wellbore instability, formation influxes, tool failures falling into contingencies, or even plugging and abandon well. In addition, the level of uncertainty in exploration wells raises a need for a system that allows quicker intervention if large variance from the prognosis and operational flexibility in terms of quick adjustment of Bottom hole Pressure (BHP) via Surface Backpressure (SBP) instead of reducing or increasing Mud Weight (MW).

Specifically, as per prognosis the main challenge of this exploration well was to drill across through high temperature environments that could potentially damage the directional tools or could compromise the quality of the logs while drilling. Additionally, there was the bottom hole pressure uncertainties crossing upper formation, immediately above the target reservoir. The well plan included a contingency scenario to isolate the high-pressure formation in case difficulties arose that prevented drilling ahead in the same hole diameter.

To address these challenges mentioned, a combined application of Managed Pressure Drilling (MPD) and Continuous Circulation System (CCS) technologies was proposed and implemented to be able to reach the Khuff formation. MPD provided real-time bottom hole pressure control while CCS ensured uninterrupted circulation during connections and tripping, minimizing thermal cycling and extending the life of downhole tools.

The main targets for this exploration well were:

- Utilize CCS and MPD in 8 ½" for a total estimated length of 7,835 ft
- Drill vertical well to a total depth (TD) of 19,300 ft
- Drill across over pressurized zone.
- Manage influx or loss scenarios before they become well control events.
- Continuous hole cleaning and solids transport
- Minimize H2S events, circulate and utilize MPD MGS and Flare Stack
- Mitigate surge and swab during tripping operations.
- Characterize formation (exploratory well)
    - Coring operations
    - Logging operations
- Avoid running the contingency 7" liner
- Promote and accomplish HSE and Service Quality standards.

Description and Application of Equipment and Processes

The CCS and MPD system were successfully rigged up, commissioned, and tested prior to drilling the 8 ½" hole section. As part of the commissioning process, fingerprinting of the surface system pressure loss was performed.

Managed Pressure Drilling

Managed Pressure Drilling (MPD) is a drilling technique that uses specialized surface equipment to actively control bottomhole pressure (BHP) during operations. Unlike conventional methods that rely solely on mud weight to balance formation pressures, MPD enables real-time adjustments by applying surface backpressure (SBP), offering precise control of BHP. For this well, the Constant Bottomhole Pressure (CBHP) method was selected to manage the upper portion of the 8½" section to cross where was expected a sudden increase in BHP. This approach maintains a stable BHP by continuously adjusting SBP during both dynamic (circulating) and static (non-circulating) conditions and effectively mitigating and reacting to formation influxes or losses.

The key component used in the MPD CBHP included:

- Rotating Control Device – closes the wellbore annulus using a passive sealing element around the drill pipe. It diverts return flow to the MPD manifold while allowing drill string rotation and vertical movement.
- Choke Manifold – regulates backpressure on the wellbore. Choke manifold in MPD is responsible for throttling the mud return in order to impose a calculated backpressure on the annulus according to the target bottom hole pressure. MPD Choke manifold is a fully automated choke manifold that is run by Programmable Logic Controllers (PLC) without human interventions. It has the capability to run in bottom hole pressure control, choke position control, surface backpressure control and standpipe pressure control.
- Coriolis Mass Flowmeter – measures real time return flow rate along with density and temperature. It serves as primary tool to identify kicks and losses. Coriolis flowmeter is very crucial part of MPD package to enable better monitoring and kick/loss detection method.
- Hydraulic Modeling and Control Software - GBSetPoint Hydraulics Simulator and Data Acquisition System (DAS) Cabin. Hydraulics modelling simulator is critical to the success of MPD operations and is an element of the primary well barrier (if statically underbalanced fluid is used) as it controls the calculation of bottom hole pressure. Any poor algorithm or less accurate model can lead to higher or less bottom hole pressure which ultimately can cause losses or kicks.
- Atmospheric gas separator (AGS): since MPD system works independent of the secondary well barrier, an MPD-dedicated, high-flow rate two-phase separator will be beneficial.
- Non-Return Valves (NRV): as per API-92M, at least two Non-Return Valves (NRVs), also called Float Valves, are required to be part of any MPD drilling string. They are vital and become elements of the primary well barrier envelope if statically underbalanced fluid is used.

MPD Process description:

The MPD system incorporated an RCD BigBore 5000, GBA MPD automatic choke Flex manifold, 4" Coriolis flow meter, pipe valves and DAS. One of the rig pumps was utilized as a back-pressure Pump and was integrated into the process to provide surface back-pressure for making MPD connections, conduct dynamic and static pore pressure tests, perform dynamic flow check and compensate Surge & Swab effects during trips out.

The RCD5000 has dual barrier rotating seal elements that seal the annulus and divert returns to the MPD choke manifold. Mud returns were taken through the 4" Coriolis mass flowmeter before being diverted either to the shakers or to the rig's mud gas separator (MGS) or MPD's mud gas separator and MPD Flare Stack. The closed drilling system enabled a significantly increased level of influx and loss detection monitoring, and automatic control of wellbore pressures while drilling (see Fig. 1— RT VND).

Before the commencement of the MPD operations, detailed hydraulics modeling was performed using GBSetpoint software. The modeling included various scenarios using different flow rates and planned mud weights used to generate the operational windows and optimize drilling parameters.

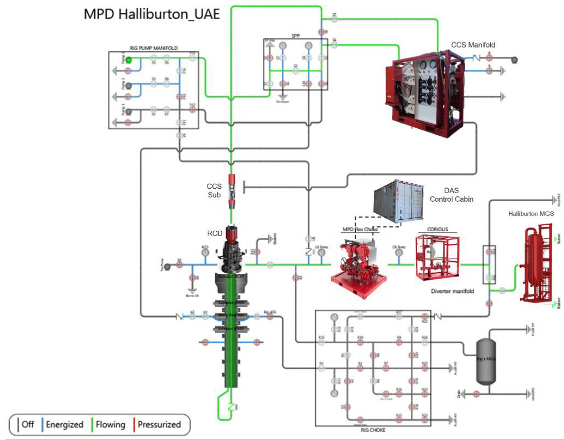

Figure 1—RT VND (Real Time Valve Numbering Diagram) shows the installation of MPD equipment and CCS on a land rig in UAE.

Continuous Circulating System (CCS)

CCS is a drilling technology designed to maintain uninterrupted circulation of drilling fluid through the drill string during pipe connections and while tripping in or out of the hole. Continuous circulation minimizes thermal cycling, thereby extending the life of the bottomhole assembly (BHA) by eliminating thermal shock to downhole tools. In addition, CCS enhances hole cleaning and reduces the risk of wellbore instability by maintaining consistent flow conditions throughout the wellbore. CCS bridges the gap between jointed pipe operation and coiled tubing operations where downhole circulation is not interrupted nor major pressure adjustment during MPD or Underbalanced Drilling (UBD) operations.

Unlike the older style of continuous circulation systems, this CCS is a manifold and a sub configuration. It consists of the following:

- CCS Circulating Subs – specialized subs that has dual flapper valve configuration, the upper flapper allows to disconnect the top drive while adding a new stand. The side flapper is used to pump drilling fluid to maintain circulation over the well while making the connection. This sub needs to be made up in every drilling stand.
- CCS Manifold – divert the flow from the rig's pump system to the Top Drive or to the side port of the sub to allow continuous circulation across the well. A hose comes out of the manifold and gets connected to the side entry valve of the circulating sub in order to establish the downhole circulation through the sub.

CCS Process description:

Before starting operations, it was necessary to pre-make up the CCS subs into stands; for this job, a total of 60 CCS subs were required. The CCS manifold must be installed next to the SPP manifold to deviate the flow from the pumps to the side port of the sub, to do that is required to install an isolation valve. Another important piece of equipment is the additional pump that must be connected to the CCS manifold to fill the new stand while circulating across the side port of the sub.

The CCS process begins once the tool joint is set on slips and the rotary table is locked. After confirming zero pressure below the safety plug, the hose is connected to the Sub's side port and the flow is diverted via CCS manifold. Once the flow is fully diverted to the side port, the axial flapper closes to isolate the flow and pressure from the SPP to the Top Drive (TDS). Using the CCS manifold pressure is safely bled off and the TDS can be disconnected, and the process continues adding a new stand. The additional pump fills the new stand with mud. Finally, the flow is diverted back from the side port to the TDS, pressure in the hose is bled off and it can be safely removed. Drilling resumes.

Presentation of Data and Results

Drilling of the 8½" section began after a conventional Formation Integrity Test (FIT) and MPD and CCS fingerprinting was performed to set a baseline on pressure and calibrate the MPD system, check pump efficiencies, verify choke controllability, understand flow behavior during cycling of the pumps on and off, determine additional drag due to the pipe movement across the RCD rubber and simulate kick and loss scenarios.

CCS fingerprinting was carried out to compare calculated pressure drops with actual measured values. For both systems, training connections were performed to familiarize all involved personnel with the operational sequence and ensure readiness before commencing drilling.

Drilling began using MPD for Early Kick Detection (EKD) and to safely circulate and flare gases during each run. The MPD system was also used to perform Dynamic Formation Integrity Tests (DFIT), Static Pore Pressure Tests (SPPT), and flow checks. During tripping operations, Surface Backpressure (SBP) was applied to minimize swab effects.

CCS connections were made continuously up to the coring depth (at Minjur and Marrat formation) and the reached the planned logging point at 15,145 ft.

However, due to elevated bottomhole temperatures, cable logging attempts were initially unsuccessful. To address this, multiple conditioning runs were performed along with MPD bearing installations and gas circulation operations. Surface backpressure of 300 psi was applied during trips to mitigate swab-induced gas influxes. Wireline logging data from these operations enabled an estimation of the bottomhole temperature increase over time (see Table A-1).

Table A-1—Logging information at 15,145 ft

Last circulation with drill string	Time of Log	time since last circulation (hrs.)	Temperature Registered (F) w/Wireline	Raised (F) between log runs	ΔTime (hrs) between runs	Temp (F) increase per hour
9/9/2024 16:00	9/12/2024 2:45	58.75	352	6	14.17	2.36
	9/12/2024 16:55	72.92	358			
9/14/2024 22:00	9/16/2024 13:00	39	340	3	7	2.33
	9/16/2024 20:00	46	343			
9/18/2024 23:30	9/20/2024 2:30	27	332			

Analysis showed a consistent temperature rise in the range of 2.33 to 2.36°F per hour, assuming a stable circulating temperature of 240°F during conditioning trips (value gathered during drilling) (see table A-2). While the average observed increase matched closely with cable log data (approximately 2.35°F/hr), the rate of heat transfer is nonlinear, gradually decreasing as the fluid temperature approaches thermal equilibrium with the surrounding formation.

Table A-2—Logging information vs circulating temperature at 15,145 ft

#	Depth (ft)	Avg circulating temp (LWD) (F)	Temperature Registered (F) w/Wireline	Raised (F) from last circ to log run	Time (hrs) from last circ to log run	Temp (F) increase per hour
1	15145	240	352	112	58.75	1.91
2	15145	240	358	118	72.92	1.62
3	15145	240	340	100	39.01	2.56
4	15145	240	343	103	46.20	2.24
5	15145	240	332	92	27.05	3.41

Drilling with MPD and CCS resumed adjusting the MW from 14.10 to 16.60 ppg, and it progressed to 16,836 ft. After two successful CCS connections, a sealing issue was encountered; the affected CCS subs were replaced and connections resumed. As per the MPD and CCS program the steps to follow after an event was to apply SBP to maintain constant bottom hole while replacing the sub. This troubleshooting process was repeated in the next connections, alternating with MPD connections as needed, until reaching 18,044 ft, at which point the string was pulled out using elevators to facilitate sub cleaning. It was identified that the sub's seal was being compromised due to the presence of dried drilling fluid residue, as the subs had remained installed and exposed to environmental conditions for more than three weeks during the attempts to perform wire logs and conditioning runs.

From 18,044 ft to the well's total depth (19,300 ft), drilling proceeded without issues. For the following wiper log run using triple combo tools, CCS was used to cool down the well. On November 16th, 37 successful CCS connections were completed while running in hole.

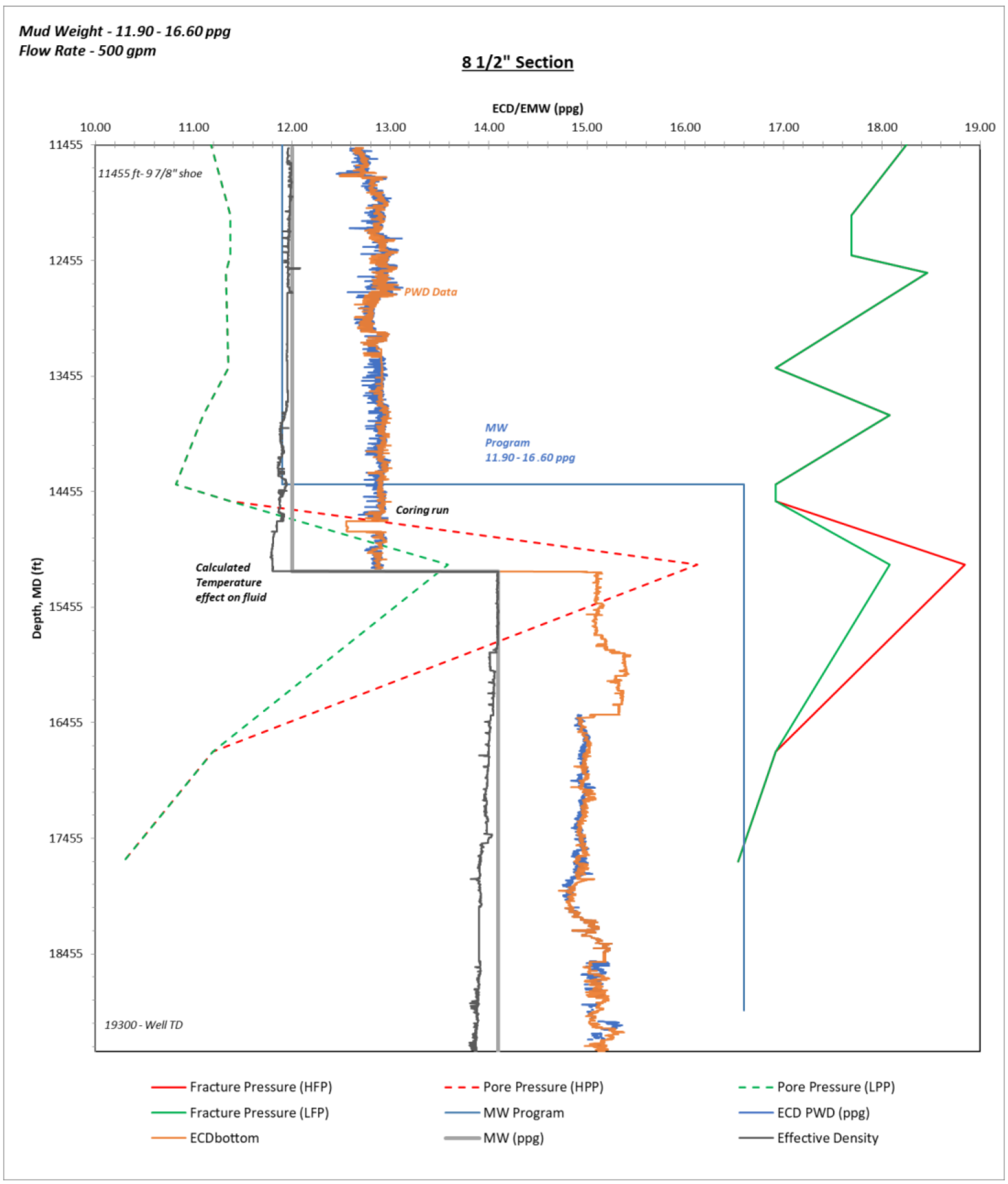

Figure 2—ECD vs Depth (MD), including planned MW vs real MW and Pore and fracture gradient as per prognosis.

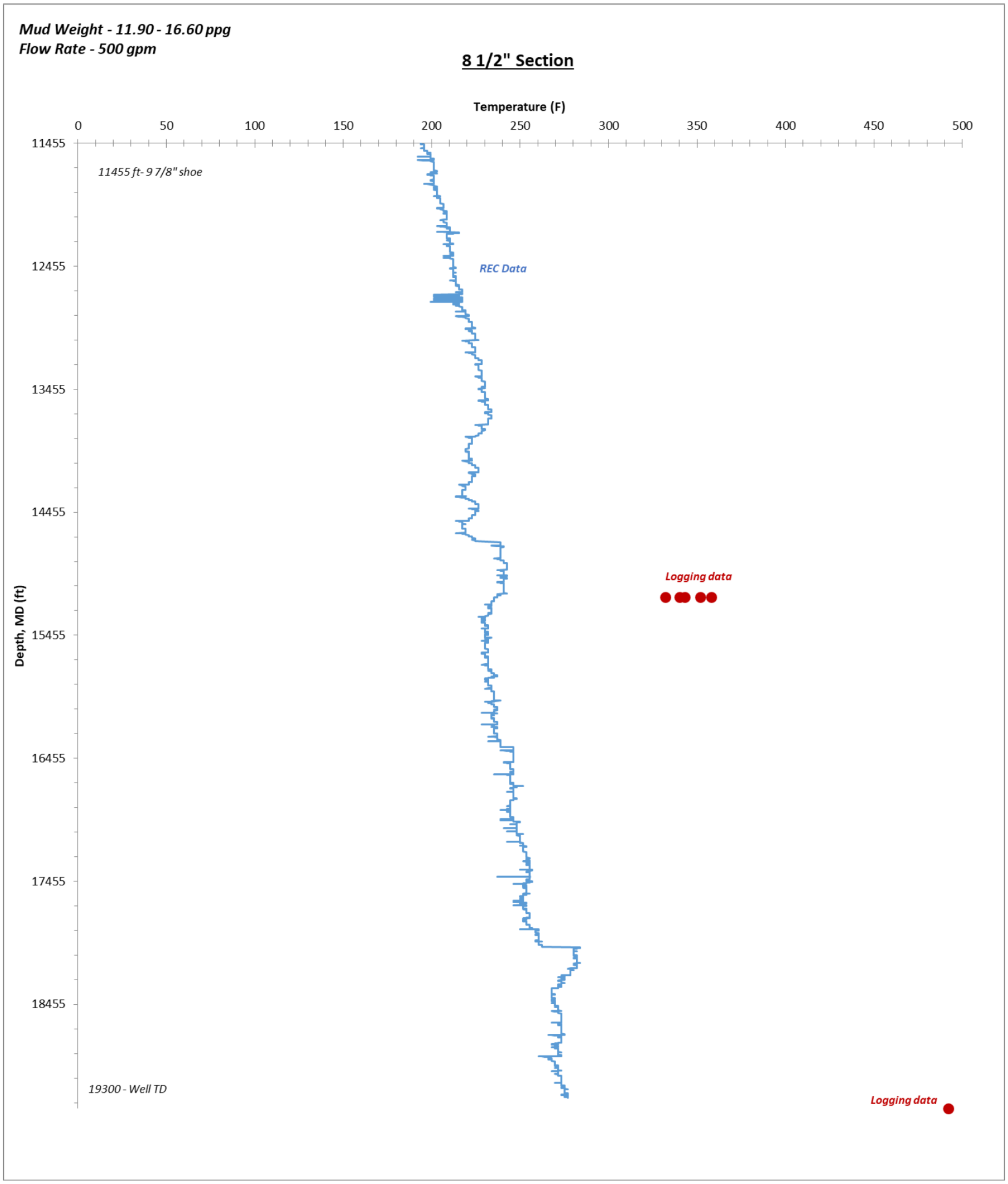

Figure 3—Depth (MD) vs Bottom hole temperature

During one of the final CCS connections, the bottomhole temperature remained stable at 260°F, while the surface temperature was recorded at 157°F. The Equivalent Circulating Density also remained nearly constant, with only a slight drop observed due to the absence of pipe rotation during the connection.

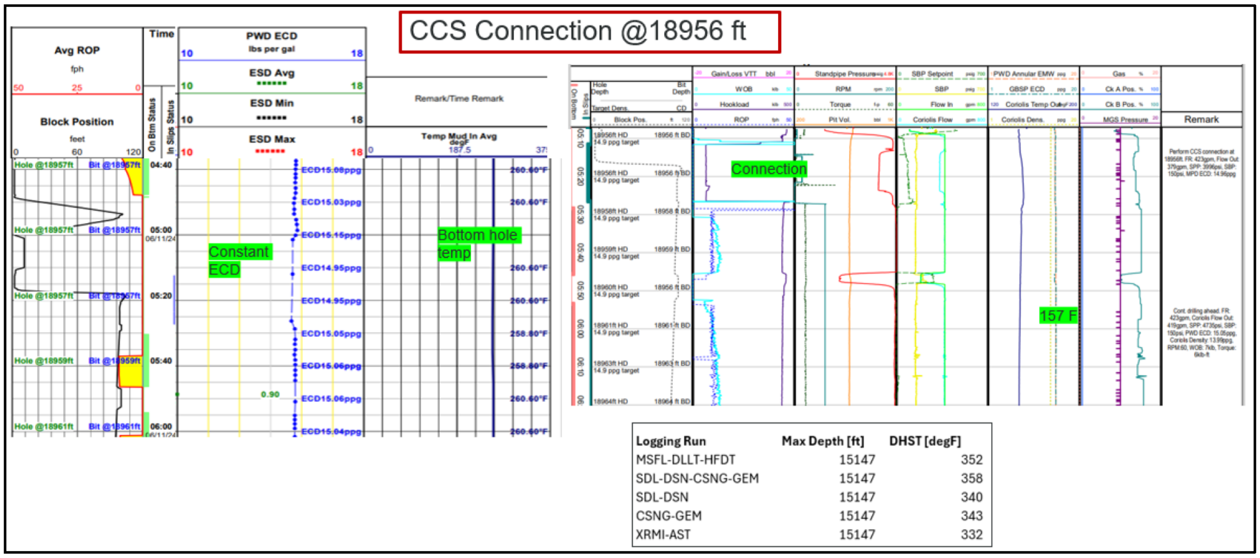

Figure 4—Example of CCS connection showcase ECD and bottom hole temperature stability

While running the Cement Bond Log (CBL), three glass thermometers were placed inside a sub, recording a bottomhole temperature of 492°F. It is important to note that an exothermic reaction occurs during cement hardening, which, according to studies, can increase the temperature by approximately 41 to 86°F, that needs to be adjusted.

Table A-3—Information recovered from glass thermometers

#	Depth (ft)	Avg circulating temp (LWD) (F)	Temperature Registered (F) w/Wireline	Raised (F) from last circ to log run	Time (hrs) from last circ to log run	Temp (F) increase per hour
6	19300	270	492	222	59	3.8

Conclusions

The combined application of CCS and MPD enabled the Operator to drill into the deep reservoirs in the Permian to successfully acquire critical formation data in an extreme high-temperature environment—without the need for a contingency liner and without facing any tool damaged. This operation marked the Operator's first use of CCS, which proved its value by enabling safe and stable drilling conditions with no kicks, losses, or signs of wellbore instability, while performing 235 successful connections during drilling and tripping. Circulating temperatures were effectively maintained around 260°F at 19,300 ft, despite static formation temperatures exceeding 490°F.

Throughout the operation, MPD and CCS worked in synergy. When CCS encountered operational challenges, MPD seamlessly took over to maintain well control and bottomhole pressure, ensuring uninterrupted progress. MPD provided reliable performance across pressurized zones, allowing continuous monitoring and adjustment of wellbore pressures. Downhole tools functioned effectively without failure; thermal shock over the directional tools was avoided with the use of the CCS.

The key objectives of implementing MPD and CCS were fully achieved, including:

- Maintaining bottomhole temperature within tool-operating limits. This was achieved via avoiding the thermal shocks that results from the downhole temperature surge when circulation is stopped to make drillpipe connection.

- Maintain ECD constant during connections and trips. This was done via the continuous circulation along with the capability to apply surface backpressure to make up for any loss of annular friction pressure.
- Enhancing hole cleaning efficiency via the uninterrupted downhole circulation.
- Performing formation characterization with MPD (use of static pore pressure and dynamic formation integrity tests became usual)
- Detecting and managing influxes: with MPD, new capabilities stem out with the continuous pressure containment achieved by Rotating Control Device (RCD) and the precise pressure control of MPD choke manifold along with the advanced kick/loss detection abilities via Coriolis flowmeter. Once influx takes place, Coriolis flowmeter is able to catch at minimal size (leading to smaller surface pressure and less downhole stresses and less risk of fracturing) where MPD system can engage by increasing Surface backpressure resulting in instantaneous bottomhole pressure increase. This will cease any flow and prepare for the next step where MPD control shifts from bottomhole pressure control to standpipe pressure control which allows circulating out the influx while maintaining constant bottomhole pressure at the new setpoint.
- Safely flaring gases through the atmospheric gas separator.
- Monitoring and controlling surface backpressure throughout operations

This case demonstrates the effectiveness of MPD and CCS technologies working in tandem to safely drill complex HPHT wells and deliver high-value formation data without compromising safety or operational integrity. Managed Pressure Drilling allowed closing the loop with Rotating Control Device (RCD) and controlling the bottomhole pressure via a fully automated choke manifold and GBSetpoint hydraulic modelling simulator which resulted in precise downhole pressure control. Continous Circulation System allowed uninterrupted downhole mud circulation which helped stabilize the downhole pressure during making connections, keeping lower downhole temperatures, improving the borehole stability and carrying the drilling cuttings for better hole cleaning and lower drag.